

A17753S1 PROJECT DISSERTATION

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Title	Are biotic interactions stage-dependent, and does stage dependence matter for predicting stochastic community dynamics?
Word Count	6,485

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Are biotic interactions stage-dependent, and does stage dependence matter for predicting stochastic community dynamics?

ABSTRACT

In most multicellular organisms, ontogenetic niche shifts modify the strength of biotic interactions as a function of the life cycle stage of the interacting individuals (e.g., juvenile, adult). However, the prevalence and importance of stage-dependent biotic interactions for the dynamics of communities exposed to environmental stochasticity remain poorly understood. Here, to test the prevalence of stage dependence in biotic interactions, I quantify pairwise stage-specific density-dependent effects on vital rates of survival, growth, and reproduction in an alpine herbaceous plant community in Colorado, USA, using long-term demographic data. Then, to quantify the impact of altered stage-dependent interactions on predicting stochastic community dynamics, I simulate two-species communities with stage-dependent competition and determine the accuracy of predictive community models with or without stage-dependent interactions. I find that (1) greater stage dependence reduces the accuracy of predictive models without stage-dependent interactions, but that (2) predictive models without stage-dependent interactions can still accurately predict community dynamics with stage-dependent interactions and that (3) stage-dependent interactions are either absent or undetectable in the alpine plant community. These findings highlight the challenges of quantifying stage-dependent interactions in empirical systems and suggest that community forecasts for conservation may not need stage-specific information in biotic interactions.

INTRODUCTION

Biotic interactions are a major driver of natural community dynamics. Here, a biotic interaction is an effect exerted by an organism on another which leads to positive or negative consequences (Morales-Castilla et al., 2015). Biotic interactions manifest as density-dependent effects, where the net biotic interaction experienced by a recipient varies with the density of actors (Murray, 1982). Density-dependent effects may influence population growth rates (Berryman et al., 2002; Murray, 1982) or vital rates, *i.e.*, rates of survival, growth, and reproduction (Neubert & Caswell, 2000). In turn, density-dependent effects drive natural community dynamics together with environmental and demographic stochasticity (Mutshinda et al., 2009). Particularly, density-dependent effects underlie the regulation of communities by competition (Hanski, 1997; Peters, 2003) or trophic interactions (Heath et al., 2014; Sibly et al., 2002; Thakur & Geisen, 2019). Anthropogenic drivers such as climate change and invasive species can alter biotic interactions, which may disrupt these regulatory mechanisms and mediate biodiversity loss (Blois et al., 2013; Waller et al., 2020). As such, community models with explicit biotic interactions may aid conservation by predicting community dynamics under specific pressures (Boult, 2023; Petchey et al., 2015).

Biotic interactions are shaped by ontogeny. In all multicellular organisms, development controls morphological, physiological, and behavioural traits that, in turn, govern interactions with the biotic and abiotic environment (Dayrell et al., 2018; Nakazawa, 2015). Such an ecological shift organised along an ontogenetic dimension (*e.g.*, age, size) is termed ontogenetic niche shift (Dayrell et al., 2018; Nakazawa, 2015). An important aspect of ontogenetic niche shift is ontogenetic variation in biotic interactions (Miller & Rudolf, 2011). That is, the magnitude and

direction of density-dependent effects can vary within a population depending on the life stage of interacting individuals. For example, adults of the desert sagebrush (*Ambrosia dumosa*) promote conspecific juvenile survival but hinder conspecific adult growth via competition (Miriti, 2006). Stage-dependent interactions have also been observed in the trophic ecology of aquatic animals (Persson & Crowder, 1998; Rudolf, 2020) and within-guild competition among insect herbivores (Karban, 1989) and herbaceous plants (Grace, 1985; Leger & Espeland, 2010; Schiffrers & Tielbörger, 2006).

When biotic interactions are stage dependent, population structures can modulate density-dependent effects and thus impact community dynamics (Miller & Rudolf, 2011) (Fig. 1A & 1B). Specifically, with stage-dependent interactions, population-averaged per-capita density-dependent effects vary with the proportional abundance of each life stage (Miller & Rudolf, 2011) (Fig. 1B). Deterministic models suggest that the influence of population structure on density-dependent interactions can have community-level consequences (Miller & Rudolf, 2011). Examples include increased community stability (de Roos, 2021), emergence of alternative stable states (Nakazawa, 2011a, 2011b), and alternative conditions for competitive coexistence with bottom-up (Loreau & Ebenhoh, 1994; Moll & Brown, 2008; Schellekens et al., 2010) or top-down regulation (Wollrab et al., 2013). In sum, stage-dependent interactions introduce population structure as an additional dimension controlling density-dependent effects alongside population abundance (Miller & Rudolf, 2011; Nakazawa, 2015).

The effect of stage-dependent interactions on the importance of population structure in modelling stochastic community dynamics remains unknown. Natural communities are constantly exposed to environmental and demographic stochasticity

(Crone, 2016; Sæther, 1997) and to aperiodic disturbances (Bazzaz, 1983). Stochasticity and disturbances cause variability in population abundance and structure, which in turn, can alter vital rates via density dependence (Sinclair & Pech, 1996) (Fig. 1C). Correspondingly, the importance of population structure in driving fluctuations in population abundances may increase with (1) stronger stage dependence (*i.e.*, greater between-stage difference between per-capita interaction strengths) and (2) greater variation in population structure. Of these factors, variation in population structure depends on the focal species' life history strategy (*i.e.*, collection of traits describing an organism's life cycle *sensu* Stearns (1992)) (Capdevila et al., 2022; Sæther et al., 2013). Where population structure is a significant driver of community dynamics due to these factors, a community forecast model without stage-dependent interactions may produce inaccurate predictions (Miller & Rudolf, 2011). However, parameterising a complex community model with stage-specific interactions would likely demand data with greater resolution and volume (Getz et al., 2018), which may not be available due to limited time and resources in a conservation context (Geary et al., 2020). Given this trade-off, identifying the conditions in which such a complex model is required *vs.* conditions in which a simpler model is adequate is critical for practical applications.

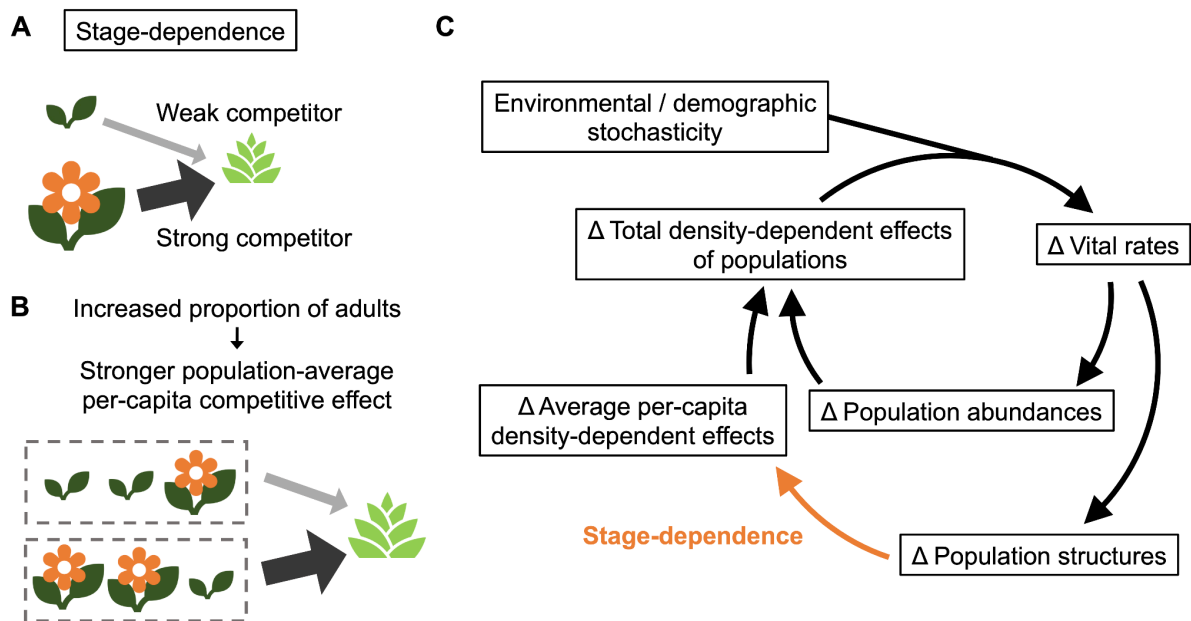


Figure 1. Stage-dependence in biotic interactions causes population-average per-capita interactions to depend on the structure of the interacting species' populations, which in turn may impact the dynamics of the whole community. (A) A hypothetical example of stage-dependent interactions. Here, the per-capita competitive effect exerted by adult plants of a species on another species is greater than that of juvenile plants of the first species. (B) In turn, stage-dependent competition generates an association between a population's structure and its average per-capita competitive effect. In this example, an increase in the proportional abundance of the more competitive adults increases the average per-capita competitive effect of the population of said species. (C) The potential role of stage-dependent interactions in modulating the stochastic dynamics of the whole community. Environmental and demographic stochasticity alter population-level vital rates of survival, growth, and reproduction, which generate fluctuations in population abundances and structures of the species in a given community. Then, population abundances and structures modulate the magnitude and direction of population-level density-dependent effects. Notably, stage-dependent interactions allow population structure to influence population-level density-dependent effects.

Here, I explore the prevalence of stage-dependent interactions and its impact on the importance of population structure in modelling stochastic dynamics (Fig. 2). Specifically, I ask (Q1) how prevalent and strong are stage-dependent interactions in a natural plant community and (Q2) how stage-dependence in biotic interactions affect the predictive accuracy of community models with or without stage dependence. To address Q1, I quantify stage-specific density-dependent effects

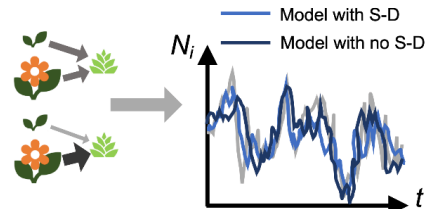
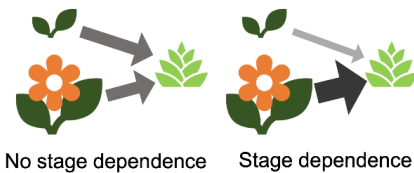
between and within three species in an alpine plant community for which demographic data exist for eight years (Ray et al., 2023). The stress-gradient hypothesis states that facilitative plant-plant interactions should be more common in more stressful abiotic conditions (Bertness & Callaway, 1994). Since alpine habitats are generally regarded as stressful to plants due to excess solar radiation and freezing (Körner, 2021), I expect that (**H1.1**) interactions between the study species will be largely facilitative. Furthermore, since larger plants often exert stronger competitive (Schwinning & Weiner, 1998) and facilitative (Miriti, 2006) influences on smaller plants, I expect that (**H1.2**) density-dependent effects of adults on juveniles will be greater than the reverse. To address Q2, I simulate stochastic two-species competition models with varying stage-dependent interactions and virtual species with linearly interpolated vital rates between extremes of the fast-slow continuum (Gaillard et al., 1989; Salguero-Gómez et al., 2016). Then, on the simulated time series, I parameterise deterministic stage-structured community models with or without stage-dependent interactions. I expect that (**H2.1**) increased stage dependence in biotic interactions will reduce the predictive accuracy of the community model without stage-dependent interactions. Since faster life histories are generally associated with greater demographic variability (Sæther et al., 2013) and greater variation in transient population growth rate (Capdevila et al., 2022), I expect that (**H2.2**) the effect of altered stage dependence will be stronger when the focal species (*i.e.*, species whose abundance is being predicted) and its competitor have faster life histories.

A**Questions****Q1**

How prevalent and strong is stage dependence in a natural community?

Q2

How does stage dependence impact the accuracy of community projection models with or without stage-dependent interactions?

**B****Methods**

Vital rate and density data from long-term demographic census of alpine plant community

Vital rates

Survival
Growth
Flowering status
Fecundity

**Local neighbour density**

Juvenile density Adult density



GLMM to quantify and compare density-dependent effects of each life stage

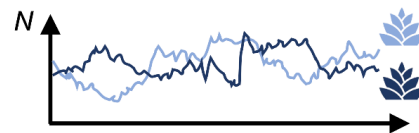
Vital rate ~

Focal juvenile neighbour density +
Focal adult neighbour density +
Other neighbour density +
Size + (1 | Plot) + (1 | Year)

Stochastic stage-structured competition model using virtual species

Virtual life history strategies

Fast life history Slow life history

**Simulation with density-dependence and stochasticity on a vital rate**

Alter stage dependence and assess accuracy of projection models with or without stage-dependent interactions

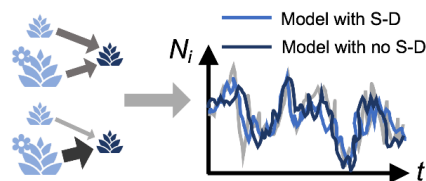


Figure 2. An overview of the questions and methods of this study. (A) Here, I ask (Q1) how prevalent and strong stage-dependent interactions are in a natural plant community and (Q2) how stage-dependence in biotic interactions affects the predictive accuracy of community models with or without stage dependence. (B) To address Q1, I quantify pairwise density-dependent effects exerted by juvenile and adult plants using a long-term demographic census dataset of an alpine plant community in Colorado. To do so, I model individual vital rates as a function of local densities of focal juveniles or adults. To quantify stage-dependence, I estimate the difference between the density-dependence effects of juveniles and adults and test its significance. To address Q2, I

construct a stochastic competition model for two virtual species with density-dependent effects and environmental stochasticity on vital rates. The virtual species span the fast-slow continuum of life history strategies. I vary the strength of stage dependence in density-dependent effects and simulate each community through time. Then, to the time series data, I parameterise community projection models with or without stage-dependent biotic interactions and compare their predictive accuracy.

METHODS

Here, I employ a combination of empirical and theoretical approaches that allow me to, together, determine the importance of stage-dependence in stochastic community dynamics. By empirically quantifying stage-dependence across a group of plant species in Q1, I assess the prominence and detectability of stage-dependence at the community level. Then, I use the empirical results from Q1 to contextualise theoretical observations in Q2, in which I quantify the potential impact of stage-dependent interactions on the accuracy of community models with or without stage dependence.

Q1: Testing for stage-dependence in plant-plant interactions

Field census

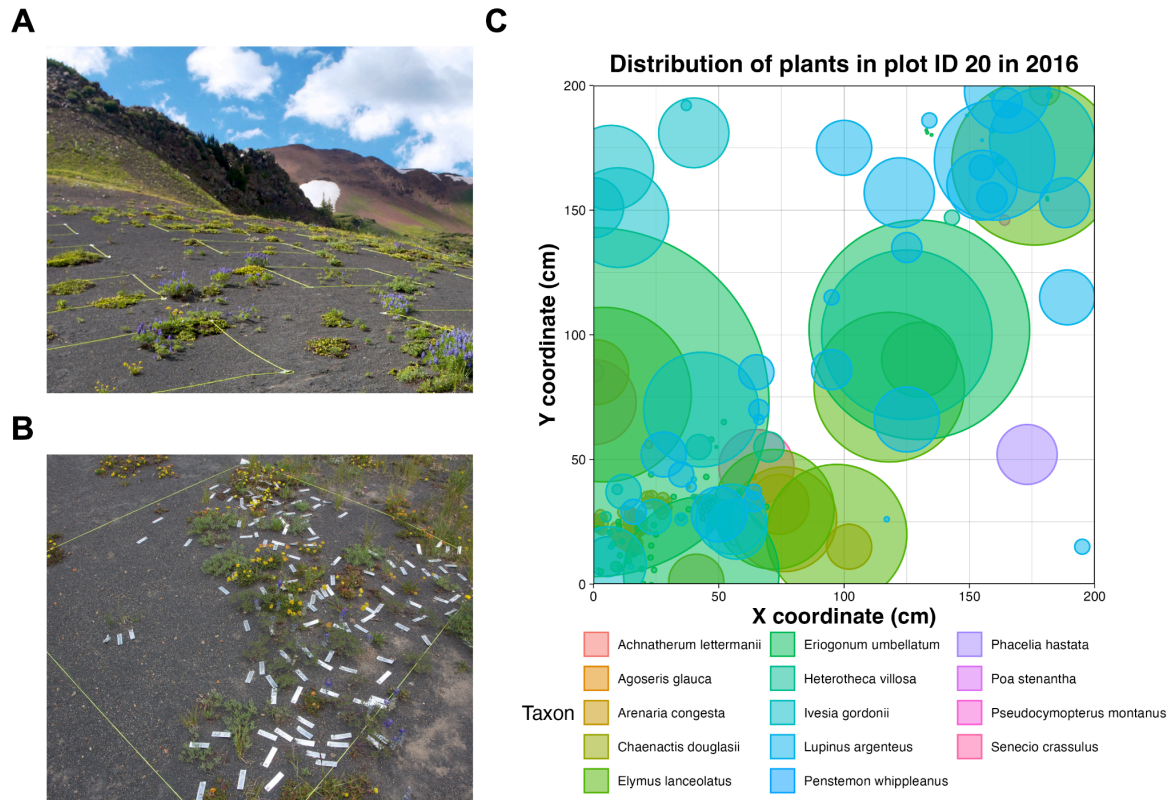


Figure 3. Field site and a visualisation of the spatially mapped census data of a natural alpine plant community in Colorado used to quantify density-dependent effects and stage dependence. (A) The field site is an alpine habitat with sparse vegetation and thin substrate above bedrock. Census plots were placed in a 5 × 10 grid as shown in the image. (B) Within each 2 m × 2 m plot, individual plants were tagged with a unique identifier. Each individual was then measured annually for the following vital rates: survival, changes in size, reproductive investment, and recruitment. (C) Visualisation of the distribution of individuals of each species in a plot. Each individual is represented with a circle, centred at the individual's position and with a radius equal to its maximum vegetative length. The colour of each circle indicates the corresponding individual's species identity. Image credit: B. Blonder.

To quantify the magnitude and direction of potential stage-dependent interactions in a natural community, I analysed an 8-year-long dataset of an alpine plant community. Censusing methods are provided in detail in Ray et al. (2023) and Blonder et al. (2018). The field site is situated on a ridge of Mt Baldy in southwestern

Colorado, United States (38.978725°N, 107.042104°W, approx. 3,540 m a.s.l.). The substrate across the site comprises thin (< 5-10 cm) gravel layered over bedrock. Vegetation is low (95% of recorded individuals $\leq$ 18 cm in height) and sparse ($\sim$ 6.0 plants m⁻² avg. in 2014). The growing season is from June to October (Inouye et al., 2000), with ground frost and snow occurring outside this window. From 2014 to 2022 (n = 8), individuals were annually censused within 50 2 × 2 m² permanent plots arranged in a 5 × 10 grid. An additional set of 50 permanent plots was installed and censused in 2022 due to a site-wide decline in plant abundance (B. Blonder & C. Ray, pers. comm., 2024). The censuses were conducted between late July and early August, coinciding with the peak of flowering of the dominant species.

All 18 observed species within this community are perennial and represent a variety of functional groups, including graminoids, forbs, and woody mat-forming plants. Since most species were rare (Fig. S1), I focused on interactions between the three most locally abundant species: *Lupinus argenteus* (Fabaceae, 20.5% of live plants across censuses), *Potentilla gordonii* (Rosaceae, 18.5%), and *Eriogonum umbellatum* (Polygonaceae, 17.0%).

For the aforementioned three focal species, I quantified individual responses to plant-plant interactions using vital rate data representing survival, growth, and reproduction. In determining survival (σ), an individual was considered alive if it produced shoot growth in the year of inspection. Environmentally controlled dormancy involving shoot dieback is a common strategy for escaping above-ground stressors in alpine plant populations (Körner, 2021). Indeed, a small proportion of individuals in the three focal species (< 0.1% of observed individuals in each focal species) were observed to be dormant for a year and subsequently produce live shoot growth. Therefore, an individual was considered dead only if it produced no

live shoot growth for two consecutive years. In determining growth (γ), I calculated the difference in the maximum shoot length of each individual between consecutive years. In quantifying reproduction, a key feature of perennial plants is that meristematic tissues may variably develop into floral or vegetative organs within an individual (Albani & Coupland, 2010; Tan & Swain, 2006). As different genetic pathways may control systemic and local flowering in perennial plants (Tan & Swain, 2006), here I considered flowering status (ϕ_0) and fecundity (ϕ_1) as separate vital rates. Fecundity was defined as the number of mature floral units (e.g., solitary flowers, capitula, umbels) and flowering status as whether an individual was flowering or not.

To quantify how individual contributions to density-dependence vary along ontogeny in a tractable manner, I categorised individuals into two discrete life stages: juveniles and adults. This simplification of the life cycle of these species is still meaningful because plant vital rates are strongly influenced by a discrete juvenile-to-adult transition (Hyun et al., 2017; Pan et al., 2023). Because size is a good predictor of vital rates in plants (Harper & White, 1974), I defined life stages based on size. Specifically, I calculated the stage threshold of each species as the greatest size of new recruits of that species observed across all censuses. Then, I assigned the adult life stage if an individual was observed as flowering in a given year t , or if its size in that year was greater than the stage threshold. By using this definition of life stages, flowering adults that were smaller than the largest new recruits were not incorrectly assigned to the juvenile stage.

Local neighbour density

To use as the explanatory variable of density-dependent effects, I calculated local neighbour density from individual size and Cartesian coordinates. To do so, I first modelled the above-ground cover $A_{i,s,k}(t)$ of each individual k in species i and life stage s (i.e., juvenile (j) or adult (a)) as a circle with a radius equal to the individual's maximum shoot length measured in years t . Circular polygons have been used to approximate individual above-ground growth in this study system, although some species exhibit uneven partial dieback (Ray et al., 2023). To exclude areas outside the plots from further analyses, I restricted above-ground cover to the area lying within plot boundaries. For each focal individual k in year t , I computed the local density $D_{k,i',s'}(t)$ of focal neighbours in species i' and life stage s' as follows:

$$D_{k,i',s'}(t) = \frac{1}{A_{i,s,k}(t)} \sum_{k'}^{N_{i',s'}} \frac{(\text{Overlapping area between } A_{i,s,k}(t) \text{ and } A_{i',s',k'}(t))}{A_{i',s',k'}(t)} \quad (\text{Eq. 1})$$

Here, $N_{i',s'}$ is the number of neighbours in the focal species and life stage. The quantity $D_{k,i',s'}(t)$ represents local density, where each neighbour's contribution is scaled continuously by overlapping above-ground cover. I chose this metric because above-ground cover reflects the spatial scale of plant-plant interactions in alpine habitats, such as sheltering from abiotic stressors (Anthelme et al., 2014; Anthelme & Dangles, 2012; Wang et al., 2016).

Statistical analyses

To test the presence of stage-dependent interactions among the focal species, I statistically quantified density-dependent effects on vital rates. To do so, I modelled the stage-specific effect of species i' (one of *L. argenteus*, *P. gordonii*, *E.*

umbellatum) on the vital rate (θ) of each individual j of species i and life stage s (i.e., juvenile or adult) using a generalised linear mixed model (GLMM):

$$\theta_{i,s,k}(t) \sim l_{i,s,k}(t) + D_{k,i',j}(t) + D_{k,i',a}(t) + D_{k,\text{Other}}(t) + (1 | t) + (1 | p) \quad (\text{Eq. 2})$$

Using the explanatory variables from year t , I modelled four types of vital rates: (1) survival (σ) from t to $t + 1$, (2) survival-dependent growth (γ) from t to $t + 1$, (3) flowering status (ϕ_0) in t , and (4) fecundity (ϕ_1) of flowering individuals in t . Because juveniles are not reproductive, I modelled flowering status (ϕ_0) and fecundity (ϕ_1) only for adults. Since these vital rates were strongly influenced by individual size (Fig. S2), I included individual size ($l_{i,s,k}(t)$) as a covariate (Eq. 2). Additionally, to isolate pairwise density-dependent effects, I also included the density of non-focal neighbours ($D_{k,\text{Other}}(t)$) as a covariate (Eq. 2). Finally, to account for temporal and spatial environmental heterogeneity, I included year (t) and plot (p) as random effects (Eq. 2). To minimise issues with model convergence, I applied a linear transformation to explanatory variables to a mean of 0 and standard deviation of 1 (Bates et al., 2015). I summarised the number of datapoints used for fitting the GLMMs in Table S1. I used a logit link function to model binary survival (σ) and flowering status (ϕ_0) and a Gaussian link function to model continuous growth (γ) and discrete fecundity (ϕ_1).

To test for between-stage differences in density-dependent effects on vital rates, I quantified the confidence intervals of stage-specific interactions. To quantify stage-dependence in each model fit, I calculated the difference between the slope estimates of focal juvenile and adult neighbour densities. Because the distribution of slope estimates was skewed, I used bootstrapping over parametric methods to determine their confidence intervals following Davison & Hinkley (1997). Specifically,

I fit each GLMM to sub-samples generated by non-parametric bootstrapping ($R = 999$) and discarded instances where the model failed to converge. To test for stage-dependent interactions, I first determined the two-tailed p -value of the test statistic while correcting for the bias from finite bootstrap sampling (Davison & Hinkley, 1997, p. 141). Then, to control the familywise error rate (FWER) across all species combinations and focal vital rates, I applied Holm's (1979) procedure at a FWER of $\alpha = 0.05$. After applying the procedure, I reported whether the null hypothesis of no stage-dependence had been rejected for each model fit. To test the significance of focal density-dependent effects, I repeated this procedure for the estimated effects of juvenile density and adult density.

Q2: Simulating the impact of stage-dependence on stochastic community dynamics

Model construction

To simulate stochastic community dynamics, I constructed a structured competition model for two virtual species. I modelled the population of each of the two species using a matrix population model (MPM), which describes discrete-time transitions and per-capita contributions across discrete life stages (Caswell, 2001). Using a similar approach to Neubert & Caswell (2000), I projected each species i 's two-stage MPM from year t to $t + 1$, thus explicitly accounting for the effects of juvenile survival ($\sigma_{j,i}(t)$), adult survival ($\sigma_{a,i}(t)$), progression from juvenile to adult ($\gamma_i(t)$), retrogression from adult to juvenile ($\rho_i(t)$), and fertility ($\phi_i(t)$) as follows:

$$\mathbf{N}_i(t+1) = \begin{bmatrix} N_{j,i}(t+1) \\ N_{a,i}(t+1) \end{bmatrix} = \begin{bmatrix} \sigma_{j,i}(t)(1 - \gamma_i(t)) & \sigma_{a,i}(t)\rho_i(t) + \phi_i(t) \\ \sigma_{j,i}(t)\gamma_i(t) & \sigma_{a,i}(t)(1 - \rho_i(t)) \end{bmatrix} \begin{bmatrix} N_{j,i}(t) \\ N_{a,i}(t) \end{bmatrix} \quad (\text{Eq. 3})$$

Here, $\mathbf{N}_i(t + 1)$ is the population vector of species i in year $t + 1$, while $N_{j,i}(t)$ and $N_{a,i}(t)$ are stage-specific abundances of juveniles and adults in year t , respectively.

This model assumes that progression and retrogression occur shortly before the census, while reproduction occurs shortly after the census. This assumption reflects a late-flowering phenology, where vegetative growth precedes flowering within a short growth season (Körner, 2021, Chapter 16.2).

To generate a community time series with density-dependence and stochasticity for Q2, I modelled vital rates as random variables controlled by density-dependent effects. Specifically, I modelled the density-dependent vital rate of each species i in year t ($\theta_i(t)$) as a transformed sum of an environmental stochasticity term $\epsilon_{\theta,i}(t)$, a baseline vital rate term $c_{\theta,i}$, and a density-dependence term $\delta_{\theta,i}(t)$ as follows:

$$\begin{aligned}\theta_i(t) &= f_{\theta}(\epsilon_{\theta,i}(t) + c_{\theta,i} + \delta_{\theta,i}(t)) \\ \epsilon_{\theta,i} &\sim \mathcal{N}(0, \sigma_{\theta}^2) \\ \hat{\theta}_i &= f_{\theta}(c_{\theta,i}) \\ \delta_{\theta,i}(t) &= w_{c,j,\theta,i}N_{j,i}(t) + w_{c,a,\theta,i}N_{a,i}(t) + w_{h,j,\theta,i}N_{j,-i}(t) + w_{h,a,\theta,i}N_{a,-i}(t) \quad \textbf{(Eq. 4)}\end{aligned}$$

Here, $f(x)$ is the vital rate-specific transformation function, σ_{θ}^2 is the constant variance of the stochasticity term, $\hat{\theta}_i$ is the baseline vital rate, and $w_{c,s,\theta,i}$ and $w_{h,s,\theta,i}$ are weights controlling the per-capita density-dependent effect of individuals in stage s (juvenile (j) or adult (a)) in the same species or the other species in this two-species virtual community, respectively. Hereafter, I use the species subscript $-i$ to refer to the competitor species i . To examine the effect of stage-dependent interactions between the two species in the dynamics of the community driven by each type of vital rate separately, I modelled all vital rates except the density-dependent vital rate as a constant:

$$\theta_i(t) = \hat{\theta}_i \text{ (Eq. 5)}$$

To restrict each vital rate to biologically and mathematically valid ranges, I used bounded identity functions as transformation functions. Specifically, I used

$$f_{\sigma_j}(x) = f_{\sigma_a}(x) = f_{\gamma}(x) = f_{\rho}(x) = \begin{cases} 0, & \text{if } x < 0 \\ x, & \text{if } 0 \leq x \leq 1 \\ 1, & \text{if } x > 1 \end{cases} \text{ (Eq. 6)}$$

to restrict σ_j , σ_a , γ , and ρ to $[0, 1]$, and used

$$f_{\phi}(x) = \begin{cases} 0, & \text{if } x < 0 \\ x, & \text{if } x \geq 1 \end{cases} \text{ (Eq. 7)}$$

to restrict ϕ to $[0, +\infty)$.

I modelled stochasticity and density-dependence as having linear and additive effects on vital rates to aid interpretation of results in relation to life history strategy. Previous density-dependent MPMS have typically scaled vital rates with a negative exponential function of population size (Jensen, 1995; Neubert & Caswell, 2000; Smouse & Weiss, 1975) or a nonlinear function analogous to a logistic growth model (Miller et al., 2002). Although these approaches prevent vital rates from taking unrealistic values, the methods cause the rate of change of vital rates with density to depend on the baseline vital rate. As such, interpreting community dynamics as the joint effect of fluctuating densities and life history-dependent metrics (e.g., sensitivity of asymptotic growth rate to changes in vital rates) may become less intuitive. To ensure that per-capita density-dependent effects depend only on the strength of density-dependence (i.e., $w_{c,s,\theta,i}$ & $w_{h,s,\theta,i}$), I modelled density-dependence as an additive term and used linear transformation functions. Following the same logic, I also modelled stochasticity as an additive term, which is analogous to some previous stochastic MPMS (Miller et al., 2002; Sykes, 1969). Below, I further describe

adjustments to density-dependent weights and stochastic variance to bound vital rates within the linear portion of the transformation function.

Model parameterisation

To generate simulated community time series data with varying levels of stage dependence, I defined baseline weights of density-dependent interactions that satisfy several criteria. First, to ensure that density-dependent effects of juveniles and adults are equal (*i.e.*, no stage dependence) in the baseline, I set the baseline weights of juveniles to be equal to the corresponding weights of adults. Second, to ensure that density-dependent effects on population growth rate are negative, I set the sign of the weights to the inverse of the sensitivity of population growth rate λ to the density-dependent vital rate (Supplementary information S1). Third, to control the relative strength of intraspecific and interspecific interactions, I set conspecific weights to be 4.5 times greater than the corresponding heterospecific weights. This ratio reflects a recent meta-analysis suggesting that intraspecific competition is, on average, four to five times stronger than interspecific competition in plant communities (Adler et al., 2018). Fourth, for symmetric competition, I set baseline weights to be constant across simulations with a particular density-dependent vital rate. Finally, to restrict vital rates within the linear portion of the transformation function (Eqs. 6 & 7), I scaled the weights with the minimum baseline value of the density-dependent vital rate across the virtual life histories.

In sum, I defined the baseline effects of conspecifics or heterospecifics of life stage s where vital rate θ is density dependent ($\hat{w}_{c,s,\theta}$, $\hat{w}_{h,s,\theta}$) as follows:

$$\begin{aligned}
\begin{bmatrix} \omega_c \\ \omega_h \end{bmatrix} &= \begin{bmatrix} 1 \\ \frac{1}{4.5} \end{bmatrix}, \quad \begin{bmatrix} \hat{\sigma}_{j,\min} \\ \hat{\sigma}_{a,\min} \\ \hat{\gamma}_{\min} \\ \hat{\rho}_{\min} \\ \hat{\phi}_{\min} \end{bmatrix} = \begin{bmatrix} 0.4 \\ 0.6 \\ 0.3 \\ 0.05 \\ 0.1 \end{bmatrix}, \quad \psi = 0.3, \quad N^* = 500, \quad \begin{bmatrix} \kappa_{\sigma j} \\ \kappa_{\sigma a} \\ \kappa_{\gamma} \\ \kappa_{\rho} \\ \kappa_{\phi} \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \\
\hat{w}_{c,j,\theta} &= \hat{w}_{c,a,\theta} = \hat{w}_{c,\theta} = \kappa_{\theta} \left(\frac{\hat{\theta}_{\min} \psi}{N^* (\omega_c + \omega_h)} \right) \\
\hat{w}_{h,j,\theta} &= \hat{w}_{h,a,\theta} = \hat{w}_{h,\theta} = \hat{w}_{c,\theta} \frac{\omega_h}{\omega_c}
\end{aligned} \tag{Eq. 8}$$

Here, ω_c and ω_h are coefficients controlling the relative strengths of conspecific and heterospecific interactions, $\theta_{\min}$ is the minimum value of baseline vital rate θ across virtual life histories (see Virtual life histories, below), ψ is a constant scaling factor for interaction weights according to the minimum vital rate, and κ_{θ} is the sign of the weights for density-dependent vital rate θ . Because baseline fertilities depend on the baseline interaction weights (see Virtual life histories, below), I assigned an arbitrary value of 0.1 to $\hat{\phi}_{\min}$. See Table S2 for details regarding the baseline interaction weights for each density-dependent vital rate.

To manipulate the degree of stage-dependent interactions from the baseline (Eq. 8), I modified the strength of density-dependent effects exerted by different life stages. To do so, I defined the stage-dependence factor r_{Δ} to control the strength of stage-dependence across all interactions in a simulation with density-dependent vital rate θ :

$$\begin{aligned}
r_{\Delta} &= \left(\frac{\partial \theta_1}{\partial N_{j,1}} - \frac{\partial \theta_1}{\partial N_{a,1}} \right) / \left(\frac{\partial \theta_1}{\partial N_{j,1}} + \frac{\partial \theta_1}{\partial N_{a,1}} \right) = \left(\frac{\partial \theta_1}{\partial N_{j,2}} - \frac{\partial \theta_1}{\partial N_{a,2}} \right) / \left(\frac{\partial \theta_1}{\partial N_{j,2}} + \frac{\partial \theta_1}{\partial N_{a,2}} \right) \\
&= \left(\frac{\partial \theta_2}{\partial N_{j,1}} - \frac{\partial \theta_2}{\partial N_{a,1}} \right) / \left(\frac{\partial \theta_2}{\partial N_{j,1}} + \frac{\partial \theta_2}{\partial N_{a,1}} \right) = \left(\frac{\partial \theta_2}{\partial N_{j,2}} - \frac{\partial \theta_2}{\partial N_{a,2}} \right) / \left(\frac{\partial \theta_2}{\partial N_{j,2}} + \frac{\partial \theta_2}{\partial N_{a,2}} \right) \\
&= \frac{w_{c,j,\theta,1} - w_{c,a,\theta,1}}{w_{c,j,\theta,1} + w_{c,a,\theta,1}} = \frac{w_{h,j,\theta,1} - w_{h,a,\theta,1}}{w_{h,j,\theta,1} + w_{h,a,\theta,1}} \\
&= \frac{w_{c,j,\theta,2} - w_{c,a,\theta,2}}{w_{c,j,\theta,2} + w_{c,a,\theta,2}} = \frac{w_{h,j,\theta,2} - w_{h,a,\theta,2}}{w_{h,j,\theta,2} + w_{h,a,\theta,2}}
\end{aligned} \tag{Eq. 9}$$

The possible values of r_{Δ} include -1 (*i.e.*, only adults exert density-dependent effects), 0 (*i.e.*, per-capita density-dependent effects of juveniles and adults are equal), and 1 (*i.e.*, only juveniles exert density-dependent effects).

In manipulating stage-dependent interactions, I ensured that population-average per-capita interaction strengths remained constant at stationary equilibrium population structures. That is, for a given equilibrium proportion of juveniles and adults in the population i ($N_{j,i}^*/N^*$ and $N_{a,i}^*/N^*$, respectively):

$$\begin{aligned} \frac{N_{j,1}^*}{N^*}w_{c,j,\theta,1} + \frac{N_{a,1}^*}{N^*}w_{c,a,\theta,1} &= \frac{N_{j,1}^*}{N^*}\hat{w}_{c,j,\theta,1} + \frac{N_{a,1}^*}{N^*}\hat{w}_{c,a,\theta,1} = \hat{w}_{c,j,\theta,1} = \hat{w}_{c,a,\theta,1} \\ \frac{N_{j,2}^*}{N^*}w_{h,j,\theta,1} + \frac{N_{a,2}^*}{N^*}w_{h,a,\theta,1} &= \frac{N_{j,2}^*}{N^*}\hat{w}_{h,j,\theta,1} + \frac{N_{a,2}^*}{N^*}\hat{w}_{h,a,\theta,1} = \hat{w}_{h,j,\theta,1} = \hat{w}_{h,a,\theta,1} \\ \frac{N_{j,2}^*}{N^*}w_{c,j,\theta,2} + \frac{N_{a,2}^*}{N^*}w_{c,a,\theta,2} &= \frac{N_{j,2}^*}{N^*}\hat{w}_{c,j,\theta,2} + \frac{N_{a,2}^*}{N^*}\hat{w}_{c,a,\theta,2} = \hat{w}_{c,j,\theta,2} = \hat{w}_{c,a,\theta,2} \\ \frac{N_{j,1}^*}{N^*}w_{h,j,\theta,2} + \frac{N_{a,1}^*}{N^*}w_{h,a,\theta,2} &= \frac{N_{j,1}^*}{N^*}\hat{w}_{h,j,\theta,2} + \frac{N_{a,1}^*}{N^*}\hat{w}_{h,a,\theta,2} = \hat{w}_{h,j,\theta,2} = \hat{w}_{h,a,\theta,2} \end{aligned} \quad (\text{Eq. 10})$$

To numerically determine stationary equilibrium stage structures, I simulated the community with baseline weights for each species pair and density-dependent vital rates. Then, I calculated the mean stage structures of each population from $t = 1,000$ years to $t = 2,000$ years and calculated the average across 10 replicate simulations. To obtain the final density-dependence weights at some r_{Δ} , I solved Eqs. 9 and 10 for the density-dependence weights as simultaneous equations.

To bound vital rates within the linear portion of the transformation function, I scaled the strength of stochasticity with the minimum value of each vital rate. Specifically, I set the standard deviation of the stochasticity term (σ_{θ}) to 10% of the minimum vital rate as summarised in Eq. 8. Thus,

$$\begin{bmatrix} \sigma_{\sigma j} \\ \sigma_{\sigma a} \\ \sigma_{\gamma} \\ \sigma_{\rho} \\ \sigma_{\phi} \end{bmatrix} = \begin{bmatrix} \hat{\sigma}_{j,\min} \\ \hat{\sigma}_{a,\min} \\ \hat{\gamma}_{\min} \\ \hat{\rho}_{\min} \\ \hat{\phi}_{\min} \end{bmatrix} \cdot 0.1 = \begin{bmatrix} 0.04 \\ 0.06 \\ 0.03 \\ 0.005 \\ 0.1 \end{bmatrix} \quad (\text{Eq. 11})$$

As a result, the effects of density and stochasticity terms on vital rates were largely linear across all simulations.

Virtual species

To test H2.2 on the joint effect of stage dependence and life history, I defined five virtual life history strategies spanning the fast-slow continuum (Salguero-Gómez et al., 2016; Stearns, 1992). A faster life history is characterised by shorter generation time (*i.e.*, average age of reproduction), lower juvenile survival, faster growth, and greater reproduction (Gaillard et al., 1989; Salguero-Gómez et al., 2016). In the present model, I represented faster life histories with lower juvenile (σ_j) and adult survival (σ_a) and retrogression (ρ) and higher progression (γ) and fertility (ϕ) (Table 1). To restrict baseline vital rates to realistic values, I bounded all four vital rates, except fertility, within the range represented in the COMPADRE plant MPM database (Salguero-Gómez et al., 2015; reported by Hernandez et al., 2024).

To test H2.2 in the absence of confounding effects, I calculated fertility to control equilibrium population sizes (Table 1). Specifically, I calculated fertility such that, when both populations are at the target equilibrium size $N^* = 500$, the asymptotic growth rate (λ) is equal to 1. As such, I derived an expression for λ in terms of vital rates and solved $\lambda = 1$ for fertility when the density-dependent vital rate θ is equal to:

$$\begin{aligned} \theta &= f_{\theta}(c_{\theta} + \delta_{\theta}^*) \\ \delta_{\theta}^* &= \hat{w}_{c,\theta}N^* + \hat{w}_{h,\theta}N^* \quad \text{(Eq. 12)} \end{aligned}$$

I provide a detailed account of this procedure in Supplementary information S2. Indeed, long-term mean population sizes were controlled between 490 and 510 across all simulations (Fig. S3).

Baseline vital rate	Life history strategy				
	F2	F1	M	S1	S2
Juvenile survival	0.400	0.525	0.650	0.775	0.900
Adult survival	0.600	0.688	0.775	0.862	0.950
Progression	0.900	0.750	0.600	0.450	0.300
Retrogression	0.050	0.062	0.075	0.088	0.100
Minimum fertility	1.132	0.754	0.490	0.283	0.109
Maximum fertility	1.629	1.125	0.809	0.561	0.344

Table 1. Summary of the baseline vital rates of the five virtual life history strategies used here to address H2.2 on the interaction between stage-dependent interactions and life history. F2 represents the fastest life history strategy, followed by F1, M, S1, and S2 in descending pace of life.

To verify that these life histories have the desired demographic properties, I constructed MPMs from the baseline vital rates. From the MPMs, I calculated generation time (*i.e.*, average age of reproductive individuals in the population) and life expectancy using age-from-stage decomposition methods for matrix population models described in Caswell (2001). Indeed, the modelled fastest life history was associated with the shortest generation time (3.2 ~ 3.4 years) and life expectancy (2.0 years). In contrast, the slowest life history had the longest generation time (8.0 ~ 9.5 years) and life expectancy (14.8 years). This pattern was preserved with the mean realised vital rates after density-dependent and stochastic effects (Table S3).

Quantifying response in stochastic community dynamics

To simulate communities with varying levels of stage dependence (r_{Δ}), I obtained time-series samples from the community model. To do so, I ran 15 sets of simulations, each testing a possible combination of the five life histories, including cases where both populations have identical baseline vital rates. I included 20

replicate subsets in each set and 11 simulations in each replicate subset, with each simulation testing a particular level of stage-dependence ($r_{\Delta} \in \{-1, -0.8, \dots, 0.8, 1\}$). Community dynamics varied greatly depending on the temporal sequence of stochasticity terms, which may obscure the effect of stage-dependence. Therefore, I used 20 sets of pre-generated stochasticity terms across all life history combinations and density-dependent vital rates. For symmetry, 10 of the 20 sets were populated by swapping the stochasticity terms between two populations. To minimise the effect of non-equilibrium initial conditions, I started each simulation from the equilibrium population size ($N^* = 500$) and numerically calculated equilibrium structures (see Model parameterisation, above). To obtain a long-term time-series sample, I sampled species and stage-specific abundances ($N_{s,i}(t)$) from $t = 600$ years to $t = 2000$ years ($n = 1400$).

To address H2.1, I fit community models with or without stage-dependent interactions to each simulated time series and quantified their predictive accuracy. To do so, I modelled the density-dependent vital rate of each species i (θ_i) using the pooled time series data for each species pair and a specific value of r_{Δ} ($n = 28000$). Specifically, I fit the following linear regression models:

$$\begin{aligned} \text{Model A: } \theta_i(t) &\sim N_1(t) + N_2(t) + \epsilon_{\theta,i}(t) \\ \text{Model AS: } \theta_i(t) &\sim N_{1,j}(t) + N_{1,a}(t) + N_{2,j}(t) + N_{2,a}(t) + \epsilon_{\theta,i}(t) \quad (\text{Eq. 13}) \end{aligned}$$

Model A represents a model without stage-dependent interaction, while Model AS represents a model where stage-dependent interactions are allowed. After determining the slope estimates, I used each model to predict community dynamics from $t_0 = 600$ years to $t_{\text{end}} = 700$ years within each replicate. To do so, I first set the initial stage-specific abundances to be equal to those at $t = 600$ years in the original simulated time-series data. Then, in each year t , I calculated the predicted

density-dependent vital rate θ of each species with the slope and intercept estimates of each model. As explanatory variables, I used the community structure in that year predicted by the focal model and the stochasticity term $\epsilon_{\theta,i}(t)$ identical to the original simulation. Finally, with the predicted density-dependent vital rate, I constructed MPMs following Eq. 3 and calculated the predicted community structure at time $t + 1$. I repeated this procedure to obtain two predicted time series data corresponding to either model A or AS for each original time series.

To quantify the accuracy of each model, I calculated the mean absolute percentage error (MAPE) of the predicted time series relative to the original time series. The MAPE for each species i was calculated as follows:

$$\text{MAPE}_i = 100\% \cdot \frac{1}{t_{\text{end}} - t_0} \sum_{t=t_0+1}^{t_{\text{end}}} \left| \frac{N'_i(t) - N_i(t)}{N_i(t)} \right| \quad (\text{Eq. 14})$$

Here, $N'_i(t)$ is the abundance of species i at time t as predicted by the model and $N_i(t)$ is the abundance of species i at time t in the original time series data.

Analysis software

I conducted all data analyses and community simulations using R (v. 4.4.1). Data manipulation and visualisation were implemented using the *tidyverse* family of R packages (v. 2.0.0) (Wickham et al., 2019). GLMMs were fit and analysed using the *lme4* (v. 1.1.35.5) (Bates et al., 2015) and *car* (v. 3.1.3) (Fox & Weisberg, 2019) packages. Demographic variables were calculated from MPMs using the *popbio* (v. 2.8) (Stubben & Milligan, 2007) and *Rage* (v. 1.6.0) (Jones et al., 2022) R packages. I carried out algebraic expansions, solved equations, and evaluated derivatives and integrals using Wolfram Mathematica (v. 14.2.0.0).

RESULTS

Q1: Testing for stage-dependence in plant-plant interactions

Overall, I found no significant support for competitive or facilitative density-dependent effects or stage-dependent interactions in the three examined herbaceous species in the Colorado alpine plant community. This result does not support the expectation for (H1.1) facilitative effects and (H1.2) greater density-dependent effects of adults. Indeed, in all 54 possible conspecific and pairwise heterospecific interactions between *Lupinus argenteus*, *Potentilla gordonii*, and *Eriogonum umbellatum*, density-dependent effects and stage-dependence were not significant (Fig. 4). A summary of the statistics of every density-dependent effect and stage-dependence statistic can be found in Table S4.

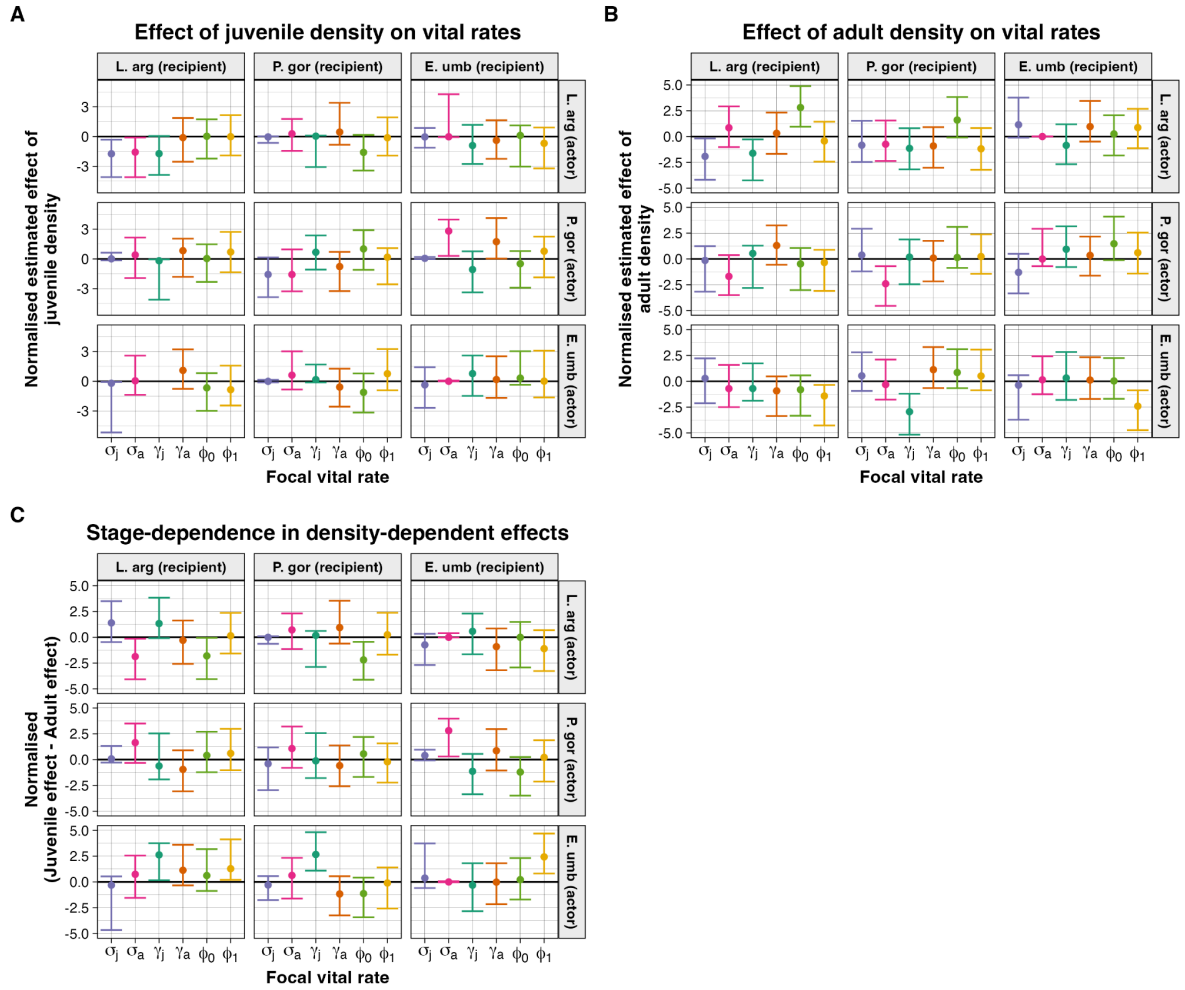


Figure 4. Density-dependent effects on vital rates or stage dependence in density-dependent effects were not significant in the empirical community. To address Q1, I quantified density-dependent effects and stage dependence in conspecific and heterospecific interactions among the three focal species (*L. agenteus*, *P. gordonii*, and *E. umbellatum*). To do so, I fit generalised linear mixed models describing vital rates in terms of stage-specific densities and obtained bootstrapped estimates for the strength of per-capita density-dependent effects from juveniles (**A**) and adults (**B**). Then, I calculated stage-dependence as the difference between the density-dependent effects exerted by juveniles and adults (**C**). The error bars indicate the bootstrapped 95% C.I. The X-axis represents focal vital rates of juvenile (σ_j) and adult (σ_a) survival, juvenile (γ_j) and adult (γ_a) growth, flowering status (ϕ_0), and fecundity (ϕ_1). Density-dependent effects and stage-dependence have been normalised to a standard deviation of 1 to aid visualisation.

Q2: Simulating the impact of stage-dependence on stochastic community dynamics

Effect of stage dependence on the predictive accuracy of community models with or without stage-dependent interactions

Consistent with H2.1, increasing the strength of stage dependence ($|r_{\Delta}|$) increased the mean absolute percentage error (MAPE) of the model without stage-dependent interactions (Model A) (Fig. 5). In contrast with H2.2, the effect of stage dependence on predictive accuracy was greatest when the focal species (*i.e.*, species whose abundance is being modelled) had the slowest life history in all four possible density-dependent vital rates except juvenile survival (σ_j) (Fig. 5). When juvenile survival (σ_j) was density dependent, faster life history of the focal species generally led to a greater MAPE of model A at a given value of stage dependence (r_{Δ}) (Fig. 5). In contrast, when progression (γ), retrogression (ρ), or fertility (ϕ) was density-dependent, the MAPE of model A at a given r_{Δ} was generally greater when the focal species had a slower life history (Fig. 5). When adult survival (σ_a) was density-dependent, there was no general trend between the pace of life of the focal species and the MAPE of model A (Fig. 5). Furthermore, the life history of the non-focal competitor species did not affect the predictive accuracy of both models A and AS (Fig. S4).

Overall, the MAPE of both models were very small (model A: < 0.7%, model AS: < 0.0003%), suggesting that both models were able to closely predict the true time series over a long timeframe of 100 years. The small MAPE of model AS is consistent with the density-dependent vital rate being a multivariate linear function of stage-specific densities and the stochastic term (Eq. 4).

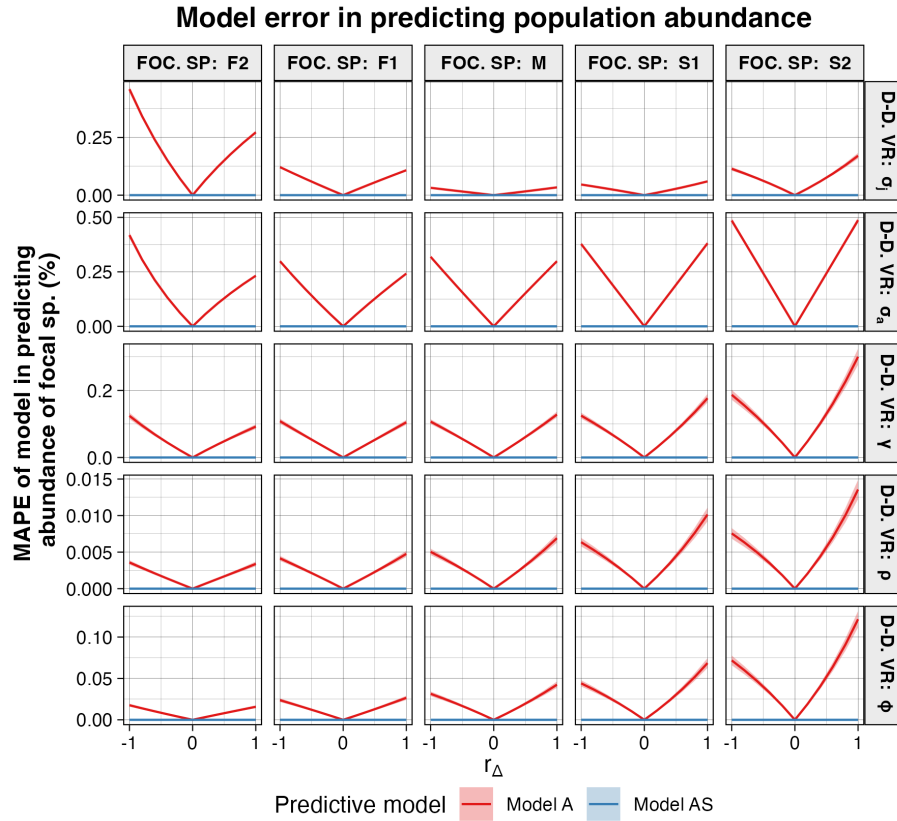


Figure 5. As expected by H2.1, the predictive error of a community model without stage-dependent interactions (Model A) increases with the strength of stage dependence, but the effect of stage dependence varies with the life history of the focal species and the density-dependent vital rate. To address Q2, I fit community projection models with (Model AS) or without (Model A) stage-dependent interactions and quantified their accuracy in predicting a 100-year simulated time series. The X-axis represents stage dependence, including simulations where only adults exert density-dependence ($r_{\Delta} = -1$), where juveniles and adults exert equal density-dependence ($r_{\Delta} = 0$), and where only juveniles exert density-dependence ($r_{\Delta} = 1$). The Y-axis represents the mean absolute percentage error (MAPE) of each predictive model in predicting the original time series data. Columns represent the life history of the focal species whose abundance is being modelled, varying between the fastest life history (F2) to the slowest life history (S2). Since the life history of the non-focal competitor species did not impact the two models' predictive accuracy (Fig. S4), the data have been pooled across different competitor identities. Rows represent the density-dependent vital rate of juvenile (σ_j) and adult (σ_a) survival, progression (γ), retrogression (ρ), and fertility (ϕ). The solid lines indicate the mean MAPE across replicates, and the 95% C.I. is shown as the shaded area.

DISCUSSION

Here, I (Q1) explored the prevalence of stage-dependent interactions in a natural community and (Q2) quantified the impact of stage dependence on the importance of stage-specific interactions in predicting stochastic community dynamics. I address a current knowledge gap in the field by modelling the joint effect of stage-dependence and stochasticity on community dynamics. First, (Q1) I build upon previous empirical work on density-dependent plant-plant interactions (Adler et al., 2018; Blonder et al., 2018; Ray et al., 2023) to model and compare density-dependent effects of juvenile and adult life stages in a natural plant community. I find that there was no support for stage-dependent interactions in the study system, stemming from the absence of statistically significant density-dependent interactions. Second, (Q2) I extend density-dependent (Jensen, 1995; Miller et al., 2002; Neubert & Caswell, 2000; Smouse & Weiss, 1975) and stochastic (Fieberg & Ellner, 2001; Nakaoka, 1996; Sykes, 1969) MPMs to determine how stage dependence influences the predictive accuracy of community models with or without stage-dependent biotic interactions. I show that stronger stage-dependence can increase the error of a community model without stage-specific biotic interactions, although this error is proportionally marginal. Together, the results from Q1 and Q2 suggest that, although stage dependence can impact stochastic community dynamics and thereby the accuracy of predictive models, (1) community models without stage-dependent interactions may be adequate for ecological forecasts and that (2) stage-dependence may be absent, weak, or undetectable at the community level in the alpine plant system I examined.

In Q1, the absence of significant density-dependent effects is contrasted with similar empirical work, which commonly detects density-dependence. Examples

include empirical studies in the focal alpine plant community (Blonder et al., 2018; Ray et al., 2023) and in grassland communities in general (Adler et al., 2018). Given that density-dependent effects have been detected within the present dataset (Blonder et al., 2018; Ray et al., 2023), I suggest that their absence in Q1 is a statistical consequence of estimating species and stage-specific density-dependent effects separately. Previous empirical studies largely partition neighbours into two categories (*i.e.*, conspecifics or heterospecifics) and estimate their respective density-dependent effects (Adler et al., 2018; Blonder et al., 2018; Chu & Adler, 2015; Ray et al., 2023). In contrast, here I partitioned neighbour densities by species and life stages, which, together with the sparse distribution of individuals, caused neighbour densities to be zero-inflated. The number of informative data points for estimating density-dependent effects was thus lower than the total sample size. This lower sample size correlates with the frequency of failed model convergence (Table S5) and may reduce statistical power (Muller, 2006).

To address these statistical issues, density manipulation experiments may be paired with observational data (Novak, 2010). An experimental approach allows density-dependent effects to be quantified across a wider range of densities compared to observational approaches (Novak, 2010). However, biotic interactions estimated via experimental methods may not align with biotic interactions as they occur in natural ecosystems. An example is Gunton & Kunin (2009), where they quantified conspecific density-dependent effects in *Silene latifolia* in experimental and observational plots. Although the experimental plot mimicked the natural variation in densities, some estimated density-dependent effects were reversed at certain scales in the natural vs. experimental plots. Therefore, density manipulation experiments may first be used to approximate a possible interaction network

between species of interest, which can then inform the statistical approaches in the observational study. For example, if density-dependent effects do not depend on the species identity of neighbours in a density manipulation experiment, observational data may be analysed using models which pool densities across species.

Furthermore, neighbour sizes may be modelled as a continuous variable instead of being partitioned into discrete stages. One approach is to define a Bayesian model to directly parameterise the relationship between neighbour size and density-dependent contribution (Barber et al., 2022; Tatsumi et al., 2016). Such methods estimate size-dependent interactions as a continuous function, which can then be used for continuous-state alternatives to MPMs such as individual-based models and integral projection models (Adler et al., 2010). Although Bayesian modelling is relatively computation-intensive, a method exists to improve performance with zero-dominated co-occurrence data (Barber et al., 2022).

The impact of stage dependence may be best explained by the amount of temporal variation in population structure. In Q2, I found that the error of a community model without stage-dependent interactions (Model A) was not always greater when the focal species had a faster life history. The difference in predictive error of model A across focal species was better explained by the amount of temporal variation in population structure. That is, focal species with greater temporal variation in population structure were associated with greater error of model A (Fig. S5). This pattern is consistent with the logical basis of H2.2, that greater deviation of population structure from its equilibrium increases the discrepancy between the true per-capita average interaction strength and the corresponding estimate without stage-dependent interactions.

In turn, the amount of temporal variation in population structure may be impacted by intrinsic features of each species' life history. Notably, the inverse relationship between the pace of life and variation in population structure seen with density-dependent progression (γ), retrogression (ρ), and fertility (ϕ) contrasts with previous empirical results (Bjørkvoll et al., 2012; Sæther et al., 2004; Sæther et al., 2013). In the present model, temporal variation in population structure may be influenced by the sensitivity of the stable population structure (*sensu* Caswell (2001)) to changes in the density-dependent vital rate. Here, I derived this quantity by differentiating an expression for the proportion of juveniles at stable population structure in terms of each of the five vital rates (Supplementary information S3) For example, when progression (γ), retrogression (ρ) or fertility (ϕ) was density-dependent, the sensitivity of the stable population structure at the realised mean vital rates was 5 to 94-fold greater in the species with the slowest life history compared to the fastest life history (Table S3). Since these virtual life histories are not reflective of real species, the generality of this pattern in natural communities cannot be verified. To address this question, the present work may be repeated using empirical vital rates which reflect ecological and evolutionary constraints on life history strategy (Bielby et al., 2007; Salguero-Gómez et al., 2016). Generally, I suggest that quantifying vital rate-specific density-dependent effects and metrics such as the sensitivity of stable population structure may reveal such complex relationships between life history and stochastic community dynamics.

The low predictive error of the community model without stage-dependent interactions is due to the large impact of environmental stochasticity and small variation in population size. Overall, the predictive error of the community model without stage-dependent interactions (model A) was extremely low ($< 0.7\%$ MAPE)

even with strong stage-dependence in the ‘true’ community model. Such marginal error of model A suggests that stage dependence was not important in explaining stochastic community dynamics in the present simulations. This result is in contrast with previous theoretical works that demonstrate the significance of stage-dependent interactions in driving community dynamics (de Roos, 2021; Loreau & Ebenhoh, 1994; Moll & Brown, 2008; Nakazawa, 2011a, 2011b; Schellekens et al., 2010; Wollrab et al., 2013). The low error of the model without stage-dependent interactions may be due to two reasons. First, the stochastic community dynamics in the current simulation model may have been largely driven by environmental stochasticity. Indeed, a prediction model with only the environmental stochasticity term also predicted the true community dynamics with a relatively low error ($< 25\%$ MAPE) (Fig. S6). Second, there was little temporal variation in population structures overall, with standard deviation in the proportion of juveniles of less than 4% of the mean (Fig. S5). To determine the generality of these two features, the present work may be replicated with varying strengths of density-dependence and environmental stochasticity.

In conservation, stage-dependent interactions may be more important in predicting stochastic community dynamics when anthropogenic drivers cause non-equilibrium population structures. Examples of such drivers with stage-dependent intensity include hunting (Torres-Porras et al., 2014), selective logging (Tsingalia, 2010), and fishing (Hsieh et al., 2010). To investigate whether the performance of the predictive models is influenced by such drivers, future work may quantify predictive accuracy following demographic disturbance. In this context, metrics of transient population dynamics (*sensu* Stott et al. (2011)) may help explain the drivers of variability in population structure across life histories. However, the

results from Q2 suggest that, when there is little fluctuation in population structure, quantifying stage-dependent interactions may not improve the accuracy of community forecasts.

In conclusion, stage-dependence in biotic interactions can impact stochastic community dynamics and its predictive modelling, although its prevalence at the community level requires further investigation. The present empirical and theoretical methodologies represent a part of a growing body of literature combining demography and community ecology (de Roos, 2021; Lytle & Tonkin, 2023; Miller & Rudolf, 2011). Particularly, approaches similar to the current model may help combine treatments of stochasticity within community ecology, such as the Modern Coexistence Theory (Chesson, 2000, 2003; Ellner et al., 2019), with life history theory including the fast-slow continuum (Gaillard et al., 1989) and beyond (Stott et al., 2024; Van de Walle et al., 2023).

ACKNOWLEDGEMENT

I am deeply grateful to Dr Rob Salguero-Gómez and Dr Christina Hernández for their supervision throughout this research project, from developing its initial ideas to the writing of this dissertation. I would also like to express my sincere gratitude to Dr Benjamin Wong Blonder and Dr Courtenay Ray for their collaboration, sharing the long-term demographic dataset used in this project and providing insight on plant interactions in the empirical system. I am grateful to all members of the SalGo Team for their warm welcome and for all the exciting conversations we have had within and outside research.

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APPENDIX

Figure S1. Taxonomic distribution of live individuals across censuses in Q1

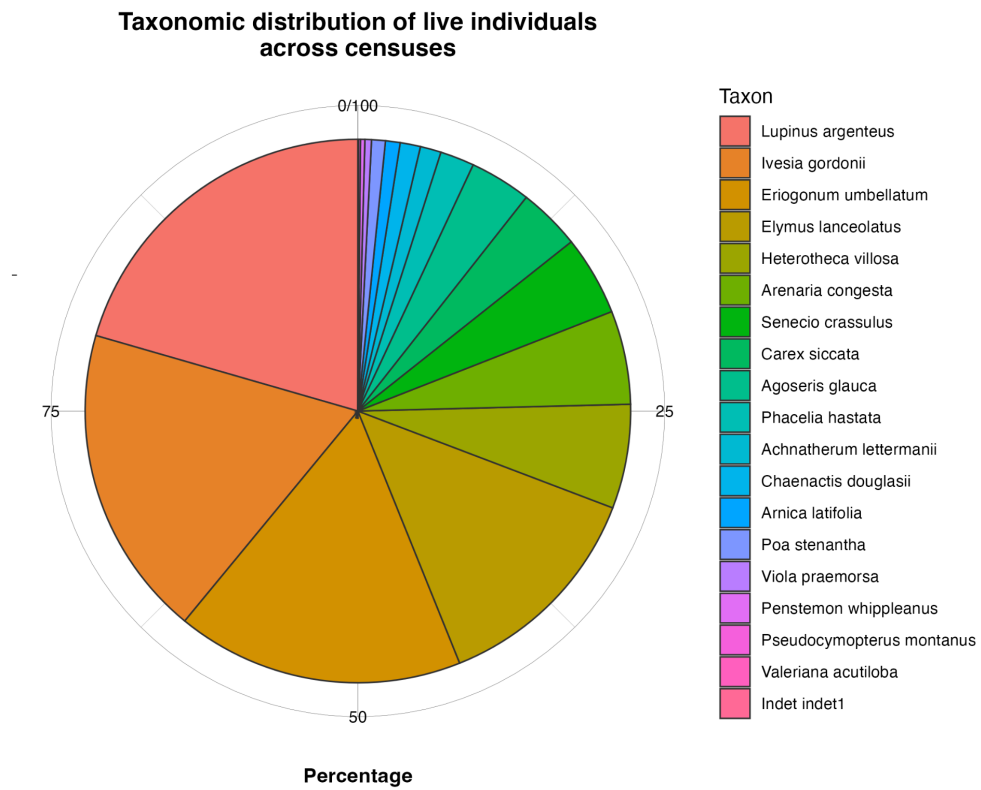


Figure S2. Plot of vital rates against individual size in focal species in Q1

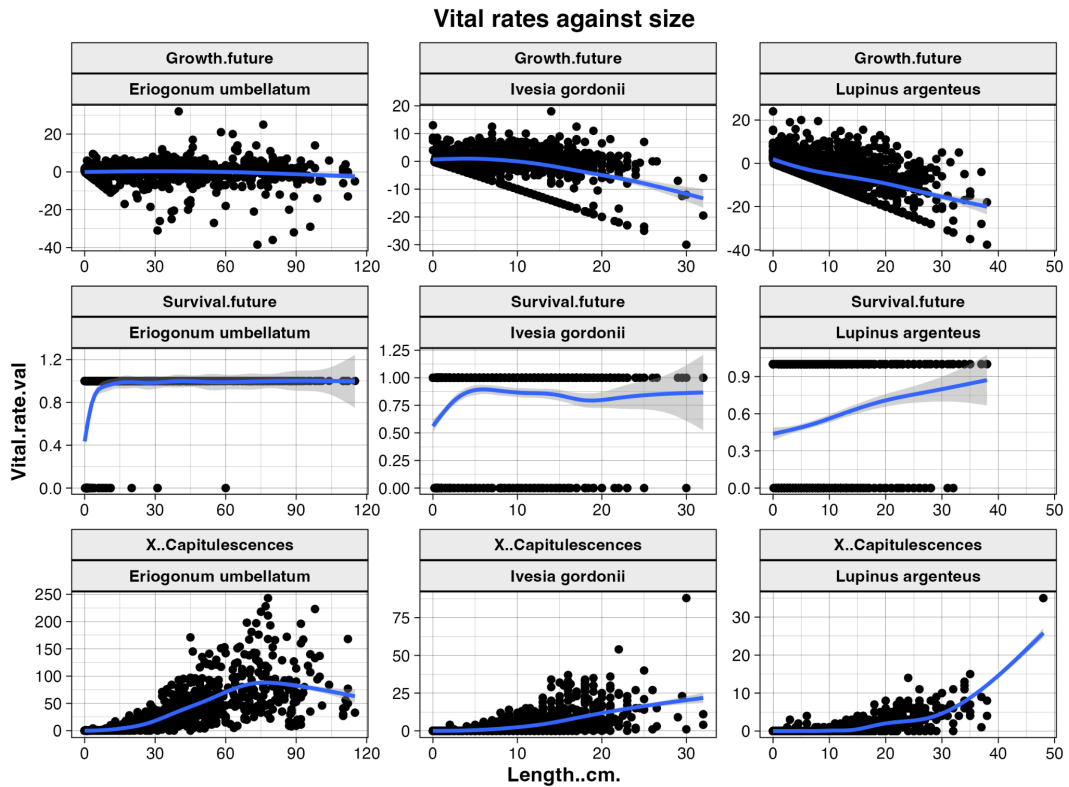


Table S1. Summary of the sample size of focal individuals in Q1 pooled across years

Focal vital rate	Focal species		
	<i>Lupinus argenteus</i>	<i>Potentilla gordonii</i>	<i>Eriogonum umbellatum</i>
Survival	n = 680 juveniles, n = 1251 adults	n = 457 juveniles, n = 1,042 adults	n = 526 juveniles, n = 874 adults
Growth	n = 299 juveniles, n = 750 adults	n = 218 juveniles, n = 985 adults	n = 288 juveniles, n = 861 adults
Flowering status	n = 1305 adults	n = 1230 adults	n = 1057 adults
Fecundity	n = 286 adults	n = 732 adults	n = 801 adults

Supplementary information S1. Sensitivities of the asymptotic population growth rate λ to each vital rate

$$\frac{\partial \lambda}{\partial \sigma_j} = \frac{1}{2} \left(\frac{2\phi\gamma + \sigma_a(\gamma\rho + \gamma + \rho - 1) + (\gamma - 1)^2\sigma_j}{\sqrt{4\sigma_j(\phi\gamma + \sigma_a(\gamma + \rho - 1)) + ((\gamma - 1)\sigma_j + (\rho - 1)\sigma_a)^2}} - \gamma + 1 \right)$$

$$\frac{\partial \lambda}{\partial \sigma_a} = \frac{1}{2} \left(\frac{\sigma_j(\gamma\rho + \gamma + \rho - 1) + (\rho - 1)^2\sigma_a}{\sqrt{4\sigma_j(\phi\gamma + \sigma_a(\gamma + \rho - 1)) + ((\gamma - 1)\sigma_j + (\rho - 1)\sigma_a)^2}} - \rho + 1 \right)$$

$$\frac{\partial \lambda}{\partial \gamma} = \frac{\sigma_j \left(-\sqrt{4\sigma_j(\phi\gamma + \sigma_a(\gamma + \rho - 1)) + ((\gamma - 1)\sigma_j + (\rho - 1)\sigma_a)^2} + 2\phi + \gamma\sigma_j + \rho\sigma_a + \sigma_a - \sigma_j \right)}{2\sqrt{4\sigma_j(\phi\gamma + \sigma_a(\gamma + \rho - 1)) + ((\gamma - 1)\sigma_j + (\rho - 1)\sigma_a)^2}}$$

$$\frac{\partial \lambda}{\partial \rho} = \frac{1}{2}\sigma_a \left(\frac{(\gamma + 1)\sigma_j + (\rho - 1)\sigma_a}{\sqrt{4\sigma_j(\phi\gamma + \sigma_a(\gamma + \rho - 1)) + ((\gamma - 1)\sigma_j + (\rho - 1)\sigma_a)^2}} - 1 \right)$$

$$\frac{\partial \lambda}{\partial \phi} = \frac{\gamma\sigma_j}{\sqrt{4\sigma_j(\phi\gamma + \sigma_a(\gamma + \rho - 1)) + ((\gamma - 1)\sigma_j + (\rho - 1)\sigma_a)^2}}$$

Table S2. Baseline interaction coefficients for each density-dependent vital rate

D-D vital rate	Species	$\hat{w}_c$	Peak $w_{c,j}$	Peak $w_{c,a}$	$\hat{w}_h$	Peak $w_{h,j}$	Peak $w_{h,a}$
σ_j	F2	-1.96e-04	-3.11e-04	-5.31e-04	-4.36e-05	-9.22e-05	-8.82e-05
σ_j	F1	-1.96e-04	-3.64e-04	-4.26e-04	-4.36e-05	-9.79e-05	-8.08e-05
σ_j	M	-1.96e-04	-4.17e-04	-3.71e-04	-4.36e-05	-1.04e-04	-7.61e-05
σ_j	S1	-1.96e-04	-4.68e-04	-3.38e-04	-4.36e-05	-1.09e-04	-7.29e-05
σ_j	S2	-1.96e-04	-5.13e-04	-3.18e-04	-4.36e-05	-1.14e-04	-7.07e-05
σ_a	F2	-2.95e-04	-4.71e-04	-7.86e-04	-6.55e-05	-1.18e-04	-1.49e-04
σ_a	F1	-2.95e-04	-5.16e-04	-6.87e-04	-6.55e-05	-1.21e-04	-1.42e-04
σ_a	M	-2.95e-04	-5.50e-04	-6.34e-04	-6.55e-05	-1.24e-04	-1.39e-04
σ_a	S1	-2.95e-04	-5.67e-04	-6.13e-04	-6.55e-05	-1.24e-04	-1.38e-04
σ_a	S2	-2.95e-04	-5.54e-04	-6.29e-04	-6.55e-05	-1.23e-04	-1.40e-04
γ	F2	-1.47e-04	-2.58e-04	-3.43e-04	-3.27e-05	-6.88e-05	-6.37e-05
γ	F1	-1.47e-04	-2.91e-04	-2.98e-04	-3.27e-05	-7.17e-05	-6.06e-05

γ	M	-1.47e-04	-3.20e-04	-2.73e-04	-3.27e-05	-7.40e-05	-5.87e-05
γ	S1	-1.47e-04	-3.40e-04	-2.60e-04	-3.27e-05	-7.55e-05	-5.78e-05
γ	S2	-1.47e-04	-3.39e-04	-2.60e-04	-3.27e-05	-7.54e-05	-5.78e-05
ρ	F2	2.45e-05	4.47e-05	5.45e-05	5.45e-06	1.25e-05	9.97e-06
ρ	F1	2.45e-05	5.10e-05	4.73e-05	5.45e-06	1.31e-05	9.44e-06
ρ	M	2.45e-05	5.70e-05	4.31e-05	5.45e-06	1.37e-05	9.08e-06
ρ	S1	2.45e-05	6.24e-05	4.04e-05	5.45e-06	1.43e-05	8.83e-06
ρ	S2	2.45e-05	6.62e-05	3.90e-05	5.45e-06	1.47e-05	8.67e-06
ϕ	F2	-4.91e-05	-9.02e-05	-1.08e-04	-1.09e-05	-2.58e-05	-1.96e-05
ϕ	F1	-4.91e-05	-1.03e-04	-9.34e-05	-1.09e-05	-2.72e-05	-1.85e-05
ϕ	M	-4.91e-05	-1.17e-04	-8.47e-05	-1.09e-05	-2.86e-05	-1.77e-05
ϕ	S1	-4.91e-05	-1.29e-04	-7.91e-05	-1.09e-05	-3.00e-05	-1.72e-05
ϕ	S2	-4.91e-05	-1.40e-04	-7.55e-05	-1.09e-05	-3.12e-05	-1.68e-05

Supplementary information S2. Procedure for calculating baseline fertility

When a vital rate other than fertility is density-dependent, I simply calculated the density-dependent vital rate at the desired equilibrium population abundances and solved $\lambda = 1$, where λ is given as:

$$\lambda = \frac{1}{2} \left(\sqrt{(\gamma\sigma_j + \rho\sigma_a - \sigma_a - \sigma_j)^2 - 4(-\phi\gamma\sigma_j - \gamma\sigma_a\sigma_j - \rho\sigma_a\sigma_j + \sigma_a\sigma_j)} - \gamma\sigma_j - \rho\sigma_a + \sigma_a + \sigma_j \right)$$

When fertility is the density-dependent vital rate, I first calculated the density-dependence term δ_ϕ at the equilibrium population sizes and solved $\lambda = 1$ when $\phi = \hat{\phi} - \delta_\phi$ for $\hat{\phi}$.

Figure S3. Mean population sizes across replicate simulations

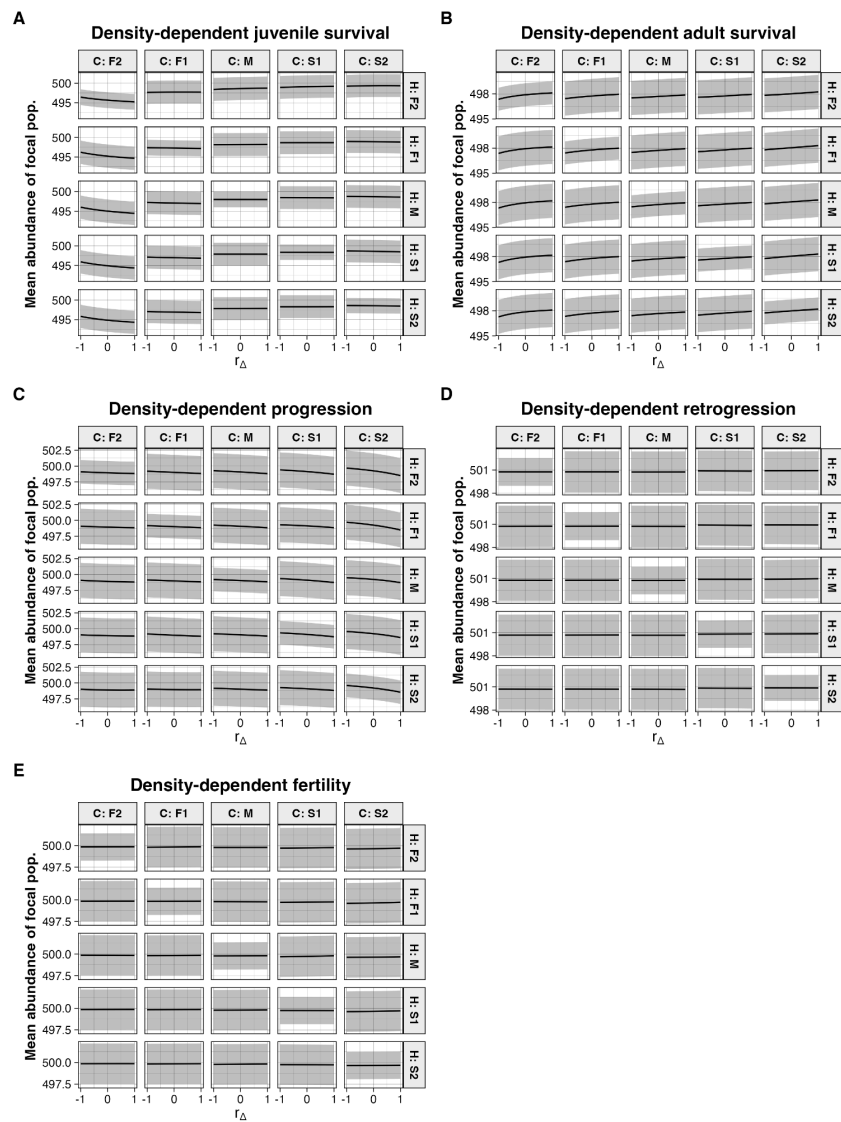


Table S4. Mean realised vital rates of each life history and corresponding demographic variables

D-D vr.	Sp.	σ_j	σ_a	γ	ρ	ϕ	Gen. time	Life expect.	Juv. prop. sens.
σ_j	F2	0.28	0.60	0.90	0.05	1.63	3.35	1.66	-1.33
σ_j	F1	0.41	0.69	0.75	0.06	1.01	3.92	2.15	-1.23

σ_j	M	0.53	0.77	0.60	0.08	0.64	4.80	2.94	-1.22
σ_j	S1	0.66	0.86	0.45	0.09	0.39	6.26	4.46	-1.31
σ_j	S2	0.78	0.95	0.30	0.10	0.19	9.10	8.70	-1.58
σ_a	F2	0.40	0.42	0.90	0.05	1.58	2.71	1.69	0.23
σ_a	F1	0.53	0.51	0.75	0.06	1.12	3.06	2.07	0.28
σ_a	M	0.65	0.60	0.60	0.08	0.81	3.58	2.67	0.33
σ_a	S1	0.78	0.68	0.45	0.09	0.56	4.40	3.72	0.39
σ_a	S2	0.90	0.77	0.30	0.10	0.34	5.97	6.23	0.43
γ	F2	0.40	0.60	0.81	0.05	1.20	3.41	1.95	0.13
γ	F1	0.53	0.69	0.66	0.06	0.80	4.03	2.53	0.23
γ	M	0.65	0.77	0.51	0.08	0.52	5.00	3.54	0.56
γ	S1	0.78	0.86	0.36	0.09	0.31	6.68	5.70	1.98
γ	S2	0.90	0.95	0.21	0.10	0.13	10.36	13.95	12.25
ρ	F2	0.40	0.60	0.90	0.06	1.13	3.32	1.96	-0.19
ρ	F1	0.53	0.69	0.75	0.08	0.75	3.88	2.56	-0.29
ρ	M	0.65	0.77	0.60	0.09	0.49	4.74	3.59	-0.43
ρ	S1	0.78	0.86	0.45	0.10	0.28	6.17	5.82	-0.65
ρ	S2	0.90	0.95	0.30	0.11	0.11	8.98	14.59	-1.13
ϕ	F2	0.40	0.60	0.90	0.05	1.12	3.37	1.97	-0.15
ϕ	F1	0.53	0.69	0.75	0.06	0.74	3.96	2.57	-0.23
ϕ	M	0.65	0.77	0.60	0.08	0.48	4.88	3.60	-0.33
ϕ	S1	0.78	0.86	0.45	0.09	0.27	6.44	5.86	-0.49
ϕ	S2	0.90	0.95	0.30	0.10	0.10	9.60	14.82	-0.82

Table S5. Summary of the median, interquartile range, p -value, and number of converged bootstrap replicates for density-dependent effects and stage-dependence

This table is available at

https://drive.google.com/file/d/13RvutSr6iP8hI069_tFIqrBoAgd5LZN2/view?usp=sharing.

Figure S4. Mean absolute percentage error of each predictive model across species combinations and density-dependent vital rates

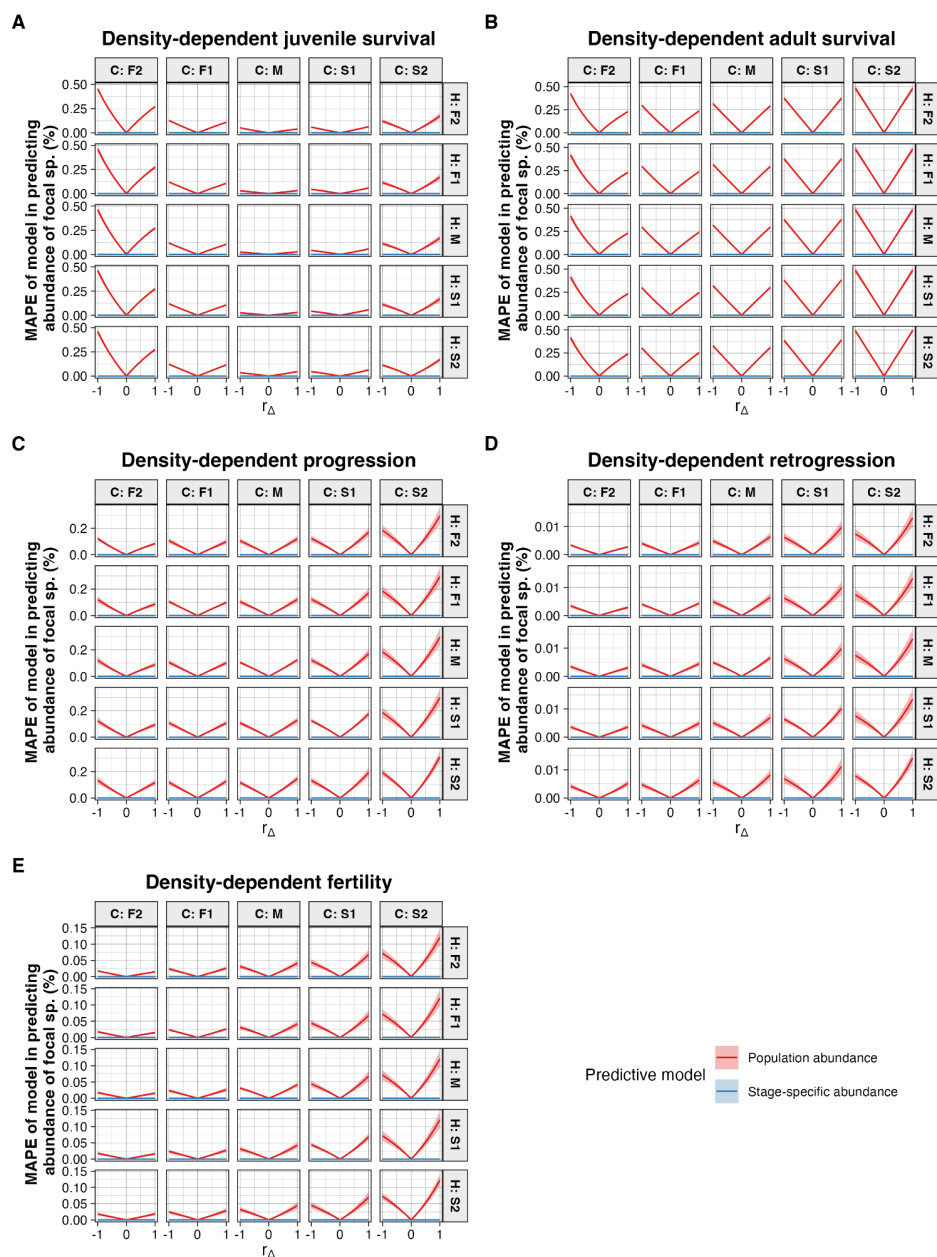
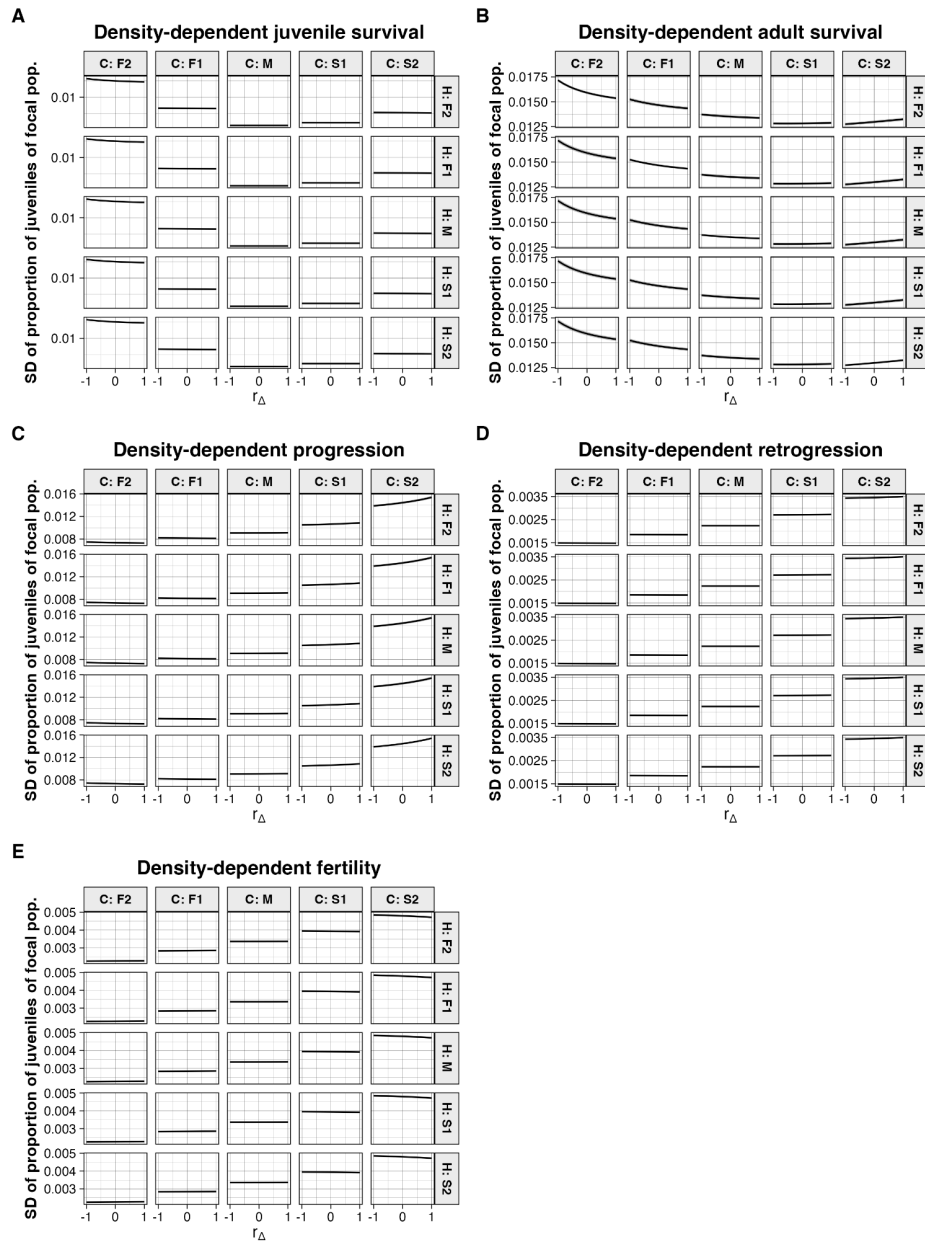
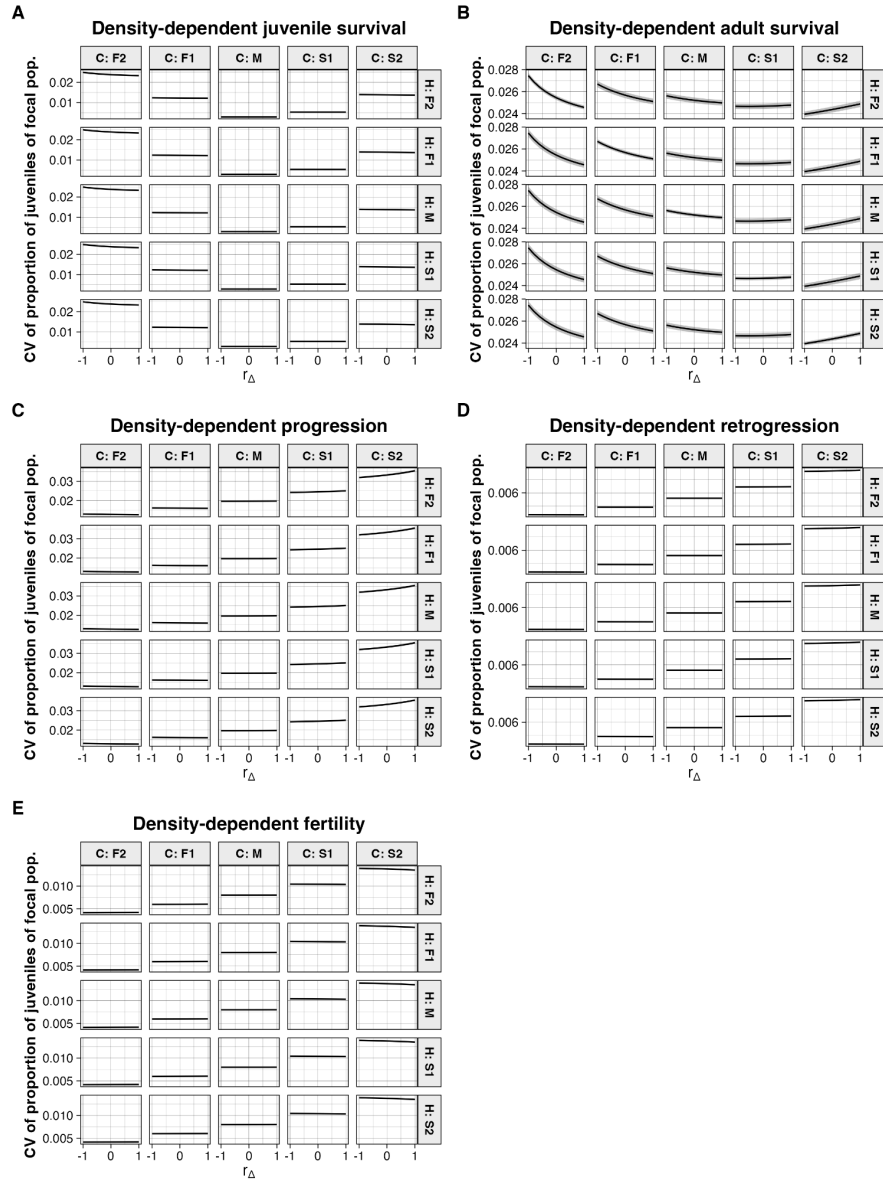


Figure S5. Standard deviation and coefficient of variance in the proportion of juveniles in each species across community simulations





Supplementary information S4. Expression for the sensitivity of the stable population structure to changes in vital rates

Given vital rates, the proportion of juveniles in the stable population structure is given as:

$$\frac{\sqrt{\sigma_j (4\phi\gamma + (\gamma - 1)^2\sigma_j) + 2\sigma_a\sigma_j(\gamma\rho + \gamma + \rho - 1) + (\rho - 1)^2\sigma_a^2} - \gamma\sigma_j + (\rho - 1)\sigma_a + \sigma_j}{\sqrt{\sigma_j (4\phi\gamma + (\gamma - 1)^2\sigma_j) + 2\sigma_a\sigma_j(\gamma\rho + \gamma + \rho - 1) + (\rho - 1)^2\sigma_a^2} + \gamma\sigma_j + (\rho - 1)\sigma_a + \sigma_j}$$

To obtain sensitivities, I differentiated this expression with respect to each vital rate.

Figure S6. Error of a predictive model with only the environmental stochasticity term

